

STAT 314500 - Applied Partial Differential Equations
Assignment 6: Green's functions

This is an individual assignment. You may collaborate with others but are expected to submit your own work. Show all relevant work. Incomplete work will be docked points.

Note: if a question is marked with an asterisk, *, and appears vague - that's on purpose. Questions marked * are conceptual short-answer questions. Do your best to reason these through, and describe that reasoning concisely. We are looking more for reasoning than precision.

Reading: Chapter 9 in Applied Partial Differential Equations with Fourier Series and Boundary Value Problems (Richard Haberman).

Question 1 Let $L = \Delta + q(x, y)$ and R be a two dimensional region. Derive an expression for (5 points)

$$\iint_R [vL(u) - uL(v)] dx dy.$$

Question 2 Let $L = \frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + q(x)$. Derive a Green's function representation for $u(x)$ satisfying (10 points)

$$\begin{aligned} Lu &= f(x) \\ u(0) &= \alpha \\ \frac{du}{dx}(L) &= \beta. \end{aligned}$$

You can assume that the Green's function for the same problem with appropriate homogeneous boundary conditions exist.

Question 3 Find the Green's function representation for the one-dimensional heat equation (10 points)

$$\begin{aligned} \frac{\partial u}{\partial t} &= k \frac{\partial^2 u}{\partial x^2} + Q(x, t) \\ u(x, 0) &= g(x) \\ u(0, t) &= 0 \\ u(L, t) &= 0. \end{aligned}$$

[Hint: Here, the Green's function should be of the form $G(x, t; x_0, t_0)$].

Question 4 Consider the one dimensional wave equation

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= c^2 \frac{\partial^2 u}{\partial x^2} + Q(x, t) \\ u(0, t) &= 0 \\ u(L, t) &= 0 \\ u(x, 0) &= f(x) \\ \frac{\partial u}{\partial t}(x, 0) &= g(x). \end{aligned}$$

Using the method of eigenfunction expansion, write a Green's function representation of the solution. (10 points)

[Hint: First verify that $h(t) = \frac{1}{\omega^2} \int_0^t g(t') \sin(\omega(t-t')) dt'$ is a particular solution to $h''(t) + \omega^2 h(t) = g(t)$.]

Question 5 Let the Green's function $G(x; x_0)$ satisfy

$$\frac{d^2 G}{dx^2} + G = \delta(x - x_0)$$

$$G(0; x_0) = 0$$

$$G(L; x_0) = 0.$$

Solve for this Green's function directly. Why is it necessary to assume $L \neq n\pi$, i.e, what happens when $L = n\pi$? Show that this Green's function is symmetric, i.e, $G(x; x_0) = G(x_0; x)$. (15 points)

Question 1

$$L = \Delta \cdot \eta(x, y) \Rightarrow \nabla L(u) - u \nabla L(v) = \nabla(\Delta u + \eta v) - u \nabla(\Delta u + \eta v) \Rightarrow \nabla L(u) - u \nabla L(v) = \nabla \Delta u - u \Delta v$$

$$\text{then } \nabla \cdot (\nabla \eta u - u \nabla \eta) = \nabla \Delta u + \nabla \eta u - u \Delta v - \nabla u \nabla v = \nabla \Delta u - u \Delta v \Rightarrow \int_{\Omega} [\nabla L(u) - u \nabla L(v)] dx dy = \int_{\Omega} \nabla \cdot (\nabla \eta u - u \nabla \eta) dx dy$$

$$\text{By divergence thm: } \int_{\Omega} [\nabla L(u) - u \nabla L(v)] dx dy = \int_{\partial \Omega} \left(\nabla \frac{\partial u}{\partial n} - u \frac{\partial \eta}{\partial n} \right) ds \quad \text{where } \frac{\partial \eta}{\partial n} = \nabla \eta \cdot n \text{ is the outward normal derivative.}$$

Question 2

$$L = \frac{1}{2} \int_{\Omega} (p(x) \frac{\partial u}{\partial x}) \cdot \eta(x), \quad Lu = f(x), \quad u(0) = \alpha, \quad u'(L) = \beta$$

$$L_u G(x; x_0) = \delta(x - x_0), \quad G(x; 0) = 0, \quad G_x(x; L) = 0. \text{ Then using the hypothesis: } G L u - u L G = \frac{1}{2} \int_{\Omega} [p(x_0) (G u - u G_{x_0})] \text{ and integrating over } 0 < x_0 < L:$$

$$\int_0^L (G(x; x_0) f(x_0) - u(x_0) \delta(x - x_0)) dx_0 = \int_{\Omega} p(x_0) (G u - u G_{x_0}) dx_0 \Rightarrow \int_0^L G(x; x_0) f(x_0) dx_0 - u(x) = p(L) G(x; L) \beta - p(0) \alpha G(x; 0)$$

$$\therefore \text{the green's function representation is } u(x) = \int_0^L G(x; x_0) f(x_0) dx_0 - p(L) \beta G(x; L) - p(0) \alpha G(x; 0) \quad \text{where } G(x, 0) = 0, \quad G_x(x, L) = 0 \text{ is a homogeneous boundary condition.}$$

Question 3

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} + O(x, t), \quad u(x, 0) = g(x), \quad u(0, t) = 0, \quad u(L, t) = 0$$

$$\text{The eigenfunctions for this BC are } u_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2$$

$$\text{then the Green's function has the form: } G(x, t; x_0, t_0) = \begin{cases} \frac{2}{L} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi x_0}{L}\right) e^{-\lambda_n(x-t-t_0)} & t > t_0 \\ 0 & t < t_0 \end{cases}$$

$$\text{Therefore, Green's function representation is } u(x, t) = \int_0^L G(x, t; x_0, 0) g(x_0) dx_0 + \int_0^t \int_0^L G(x, t; x_0, t_0) O(x, t_0) dx_0 dt_0 \quad \text{where } \lambda_n = \left(\frac{n\pi}{L}\right)^2$$

Question 4

$$u_t = c^2 u_{xx} + Q(x, t), \quad u(0, t) = u(L, t) = 0, \quad u(x, 0) = f(x), \quad u_t(x, 0) = g(x)$$

$$\text{using the same eigenfunctions: } u_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad \omega_n = c \sqrt{\lambda_n}$$

$$\text{take } u(x, t) = \sum_{n=1}^{\infty} u_n(t) \sin\left(\frac{n\pi x}{L}\right) \text{ then } u_n'' + \omega_n^2 u_n = Q_n(t) \text{ and we have } f_n = \frac{2}{L} \int_0^L f(x_0) \sin\left(\frac{n\pi x_0}{L}\right) dx_0, \quad g_n = \frac{2}{L} \int_0^L g(x_0) \sin\left(\frac{n\pi x_0}{L}\right) dx_0 \text{ and } Q_n(t) = \frac{2}{L} \int_0^L Q(x_0, t) \sin\left(\frac{n\pi x_0}{L}\right) dx_0$$

$$\text{then the forced oscillator equation gives: } u_n(t) = f_n \cos(\omega_n t) + \frac{g_n}{\omega_n} \sin(\omega_n t) + \frac{1}{\omega_n} \int_0^t Q_n(t_0) \sin(\omega_n(t-t_0)) dt_0$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} \left[f_n \cos(\omega_n t) + \frac{g_n}{\omega_n} \sin(\omega_n t) + \frac{1}{\omega_n} \int_0^t Q_n(t_0) \sin(\omega_n(t-t_0)) dt_0 \right] \sin\left(\frac{n\pi x}{L}\right) \text{ then with the green's function } G(x, t; x_0, t_0) = H(t-t_0) \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi x_0}{L}\right) \frac{\sin(\omega_n(t-t_0))}{\omega_n}$$

$$\text{then } u(x, t) = \int_0^L G(x, t; x_0, 0) f(x_0) dx_0 + \int_0^t \int_0^L G(x, t; x_0, t_0) g(x_0) dx_0 dt_0 + \int_0^t \int_0^L G(x, t; x_0, t_0) Q(x_0, t_0) dx_0 dt_0$$

Question 5

$$G'' + G = \delta(x - x_0), \quad G(0; x_0) = 0, \quad G(L; x_0) = 0 \text{ and for } x \neq x_0, \quad G'' G = 0 \Rightarrow G \text{ has the form } G(x; x_0) = \begin{cases} A \sin x & 0 < x < x_0 \\ B \sin(L-x) & x_0 < x < L \end{cases} \text{ and this satisfies } G(0; x_0) = 0, \quad G(L; x_0) = 0$$

$$\text{and continuity of } x = x_0 \text{ gives } A \sin x_0 = B \sin(L-x_0)$$

$$\text{The delta function gives condition } G'(x_0; x_0) - G'(x_0; x_0) = 1 \text{ and because } G'(x; x_0) = A \cos x, \quad G'(x; x_0) = -B \cos(L-x_0) \text{ we get } -B \cos(L-x_0) - A \cos x_0 = 1$$

$$\text{Then solving these equations } \Rightarrow G(x; x_0) = -\frac{1}{\sin L} \begin{cases} \sin x \sin(L-x_0), & 0 < x < x_0 \\ \sin x_0 \sin(L-x), & x_0 < x < L \end{cases} \text{ so we can write } G(x; x_0) = -\frac{\sin(\min(x, x_0)) \sin(L - \max(x, x_0))}{\sin L}$$

$$\text{But we need } L \neq \pi \text{ because } \sin L = 0 \text{ would hurt our formula. Also when } L = \pi \text{ the problem } y'' + y = 0, \quad y(0) = y(L) = 0 \text{ has so } y(x) = \sin(x) \text{ which means the operator is}$$

$$\text{not invertible and the green's function DNE.}$$

$$\text{Finally, the formula is symmetric because } \min(x, x_0) = \min(x_0, x) \text{ and likewise for max } \Rightarrow G(x; x_0) = G(x_0, x)$$